

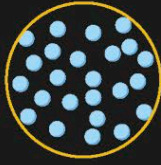
STATES OF MATTER

There are three states that all matter can exist in: solid, liquid, and gas. The ice in a stupa is a solid, but the same substance can also be water (a liquid) and vapor (gas). Solids have a fixed shape, because their molecules are held close together by strong bonds. Liquids and gases have weaker bonds, meaning liquids can change shape easily and gases have almost no shape. Ice stupas rely on the fact that liquid water can freeze into solid ice at a low temperature and ice can melt back into water at a higher temperature.



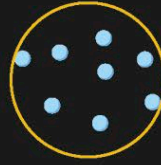
Solid

The molecules in ice are tightly packed together.



Liquid

The molecules in water have room to move.

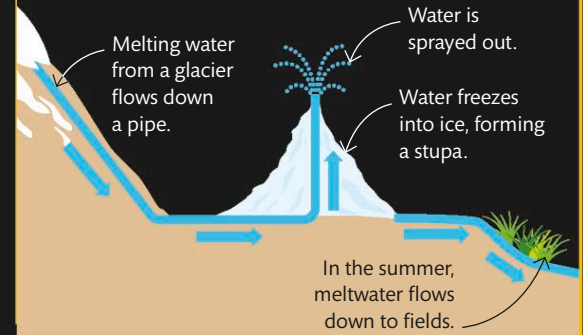


Gas

The molecules in water vapor move freely.

Making an ice stupa

Engineers lay pipes up the mountain to collect water from glaciers. The flowing water is then fed up a tube and sprayed into the freezing air, causing it to turn into solid ice. As the temperature rises, the stupa melts.



A 70-ft
(21-m) tall ice
stupa can hold
500,000 gallons
(1.9 million liters)
of water.

Water spraying
from the top
freezes into ice
in the cold air.

